

Serving = four concerns, not a stack

orthogonal decisions - they interact, but none sits on top of another

Request stream
(text → tokens)

THE FOUR SERVING CONCERNS

REUSE

prefix cache: reuse
KV of shared context

re-prefill FLOPs skipped

REQUEST SCHEDULING

continuous batching:
admit/evict each step

latency ↔ utilization

RESOURCE SPECIALISATION

P/D split: prefill pool
+ decode pool

two regimes → two pools

all four in parallel

STATE MANAGEMENT

KV cache in fixed pages
(PagedAttention)

memory capacity → concurrency

Output tokens